



Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch : III Semester B.E CSE
Course Code & Title : CS3352 & Foundations of Data Science
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** III / Describing Relationships / Standard error of estimate
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TL07
- **Activity Chosen:** Learning by doing
- **Justification**
 - ❖ Learning by Doing is an educational approach to problem-based learning. Problems on regression and Correlation is the important topic in unit III and students need to have clear idea for the conversion.
 - ❖ The standard error of the estimate shows us how the mean changes with different values. The students more familiar in the calculation, the practice problems were given in the class. The students solved themselves and discuss the same to in front of the class. It helps students in making concepts more interesting and motivating them to learn more about a specific topic.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
 - ❖ The students were informed about the event, and related problems solved in advance so that they could contribute enough time to learning the subject.
 - ❖ Instructor explained the important of Learning by Doing to the students and informed the students about it is an educational approach to problem-based learning.
 - ❖ The formulas for calculating the correction values and standard error estimate value are discussed with the students prior to the activity.
 - ❖ A student was chosen at random to solve the problem on the board using any of the teaching aids.
 - ❖ Students of Ms.M.Pooja Sree, Ms. G.Aishwarya and Ms. P.Harini . We revisited the formula steps and then students collaborated with their team mates to solve the newly assigned problem.
 - ❖ Ms. M.Pooja Sree of this class solved the problem on correlation and explained the step by step solution to entire class.

- ❖ Ms. G.Aishwarya and Ms. P.Harini solved the problem of computation formula using regression.
- ❖ Students were asked to interact and share their ideas in calculating the error estimate methods.
- ❖ Finally, the instructor consolidated the information that was discussed in this activity.
- ❖ This assisted the students to learn self and answering questions about the topic with ease.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO1
CO1	3	3	3	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovation Practice	PO1	PO2	PO3	PO4	PO12	PSO1
	2	2	1	1	1	2
Justification for Correlation	Apply basic Knowledge and fundamentals of correlation and regression techniques for solving practical problems	Identify the need of correlation and regression techniques for developing applications	Able to design solutions for applications implementing correlation and regression techniques and find relation between data	Able to design solutions for applications implementing correlation and regression techniques and find relation between data	Ability to recognize the need for correlation and regression techniques in real world applications and ability to analyze the relationship between data.	Exhibit knowledge in correlation and regression techniques for software development.

- **Images / Screenshot of the practice:**



Figure: Students actively solving problems on the class

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students learned a few things individually that leads to self-learning. They also actively involved in this session rather than it being a one-way communication.
- This activity encouraged the students to share their knowledge with others.

- ❖ ***Benefit of the practice:***

- Students successfully linked ideas through discussions, enhancing their ability to recall and understand concepts more easily
- It enhances understanding and retention of the concepts.

- ❖ ***Challenges faced in implementation***

- Slow learners were not comfortable in solving because of their individual slow learning background. However, they understood the concepts through the interaction within the team.

References:

- ❖ <https://opentextbc.ca/teachinginadigitalage/chapter/4-4-models-for-teaching-bydoing>
- ❖ https://www.ritrjpm.ac.in/images/computerscience/33_OIT551_Learningbydoing.pdf
- ❖ <https://k12.thoughtfullearning.com/blogpost/teaching-innovation-and-problem-solving>

Signature of Faculty Member

HOD